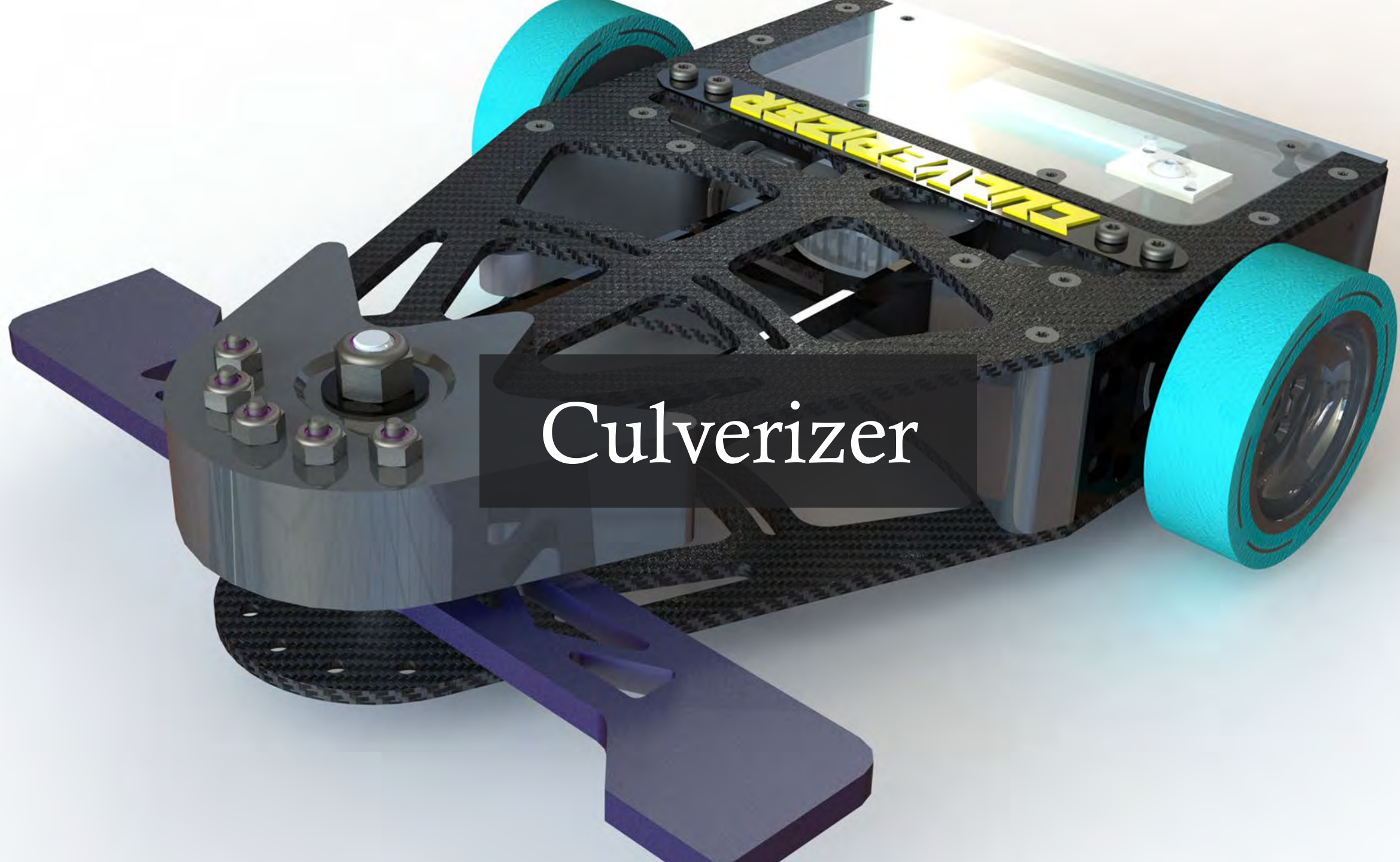


Manan Singh

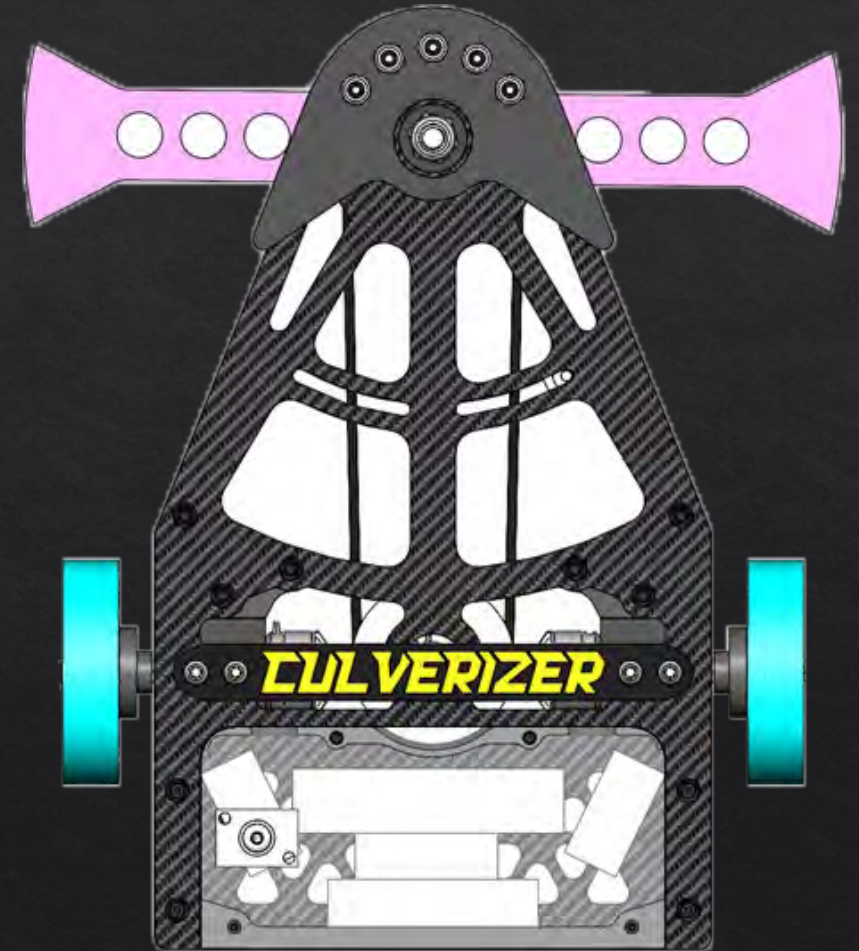
Engineering Project Portfolio



Culverizer

Culverizer – 31lb Combat Robot

- ◆ I designed and built Culverizer, which utilizes a heavy horizontal blade to batter and disable its opponents
- ◆ Carbon fiber plates give the chassis rigidity and hold weapon assembly
- ◆ 3D printed 95A TPU body flexes to absorb and disperse impact energy
- ◆ The TPU acts as ablative armor, designed to take damage and protect inner electronics



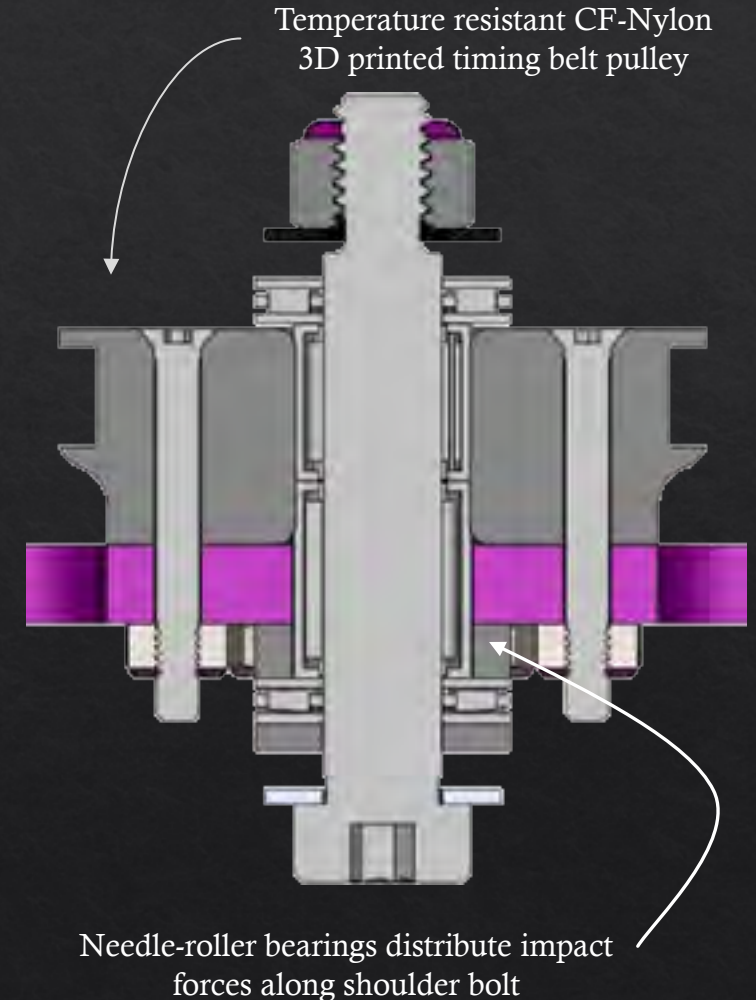
Culverizer – Drivetrain

- ◆ I used 2 brushless gearmotors to direct-drive the robot at 10.5 ft/s, enabling the bot to box-rush and engage opponents immediately
- ◆ The drive motors are shock-mounted with TPU, allowing drive motors to deflect and redirect energy away from the gearbox, preserving its lifespan
- ◆ Created custom molds to cast 2.5" OD TPU-core wheels with 40A Urethane treads
 - ◆ Urethane provides high traction and acceleration, while adhering to the TPU well
 - ◆ The TPU core absorbs impact energy and increases wheel and gearmotor durability



Culverizer – Weapon System

- ◆ 1400 kv motor @ 14.8V with a 6:5 belt reduction spins weapon blade at 340 mph
- ◆ Weapon blade stores a max of 1300J of energy
- ◆ While this iteration of the weapon system has been the most effective, I'm currently working on improvements to increase its reliability and energy transfer
 - ◆ Next iteration will replace 3D printed pulley with aluminum V-belt pulley, allowing the weapon system to slip on high impacts and reduce angular impulse delivered to the weapon motor
 - ◆ Aluminum pulley will transfer torque through a spline to avoid putting screws under shear forces



Culverizer - Competition

- ◆ Built 2 copies to be prepared for tight turnaround times between matches
- ◆ Competed at 3 tournaments with an all-time match record of 15-8
- ◆ Watch it in action! -> [Culverizer vs Surgical Procedure](#)





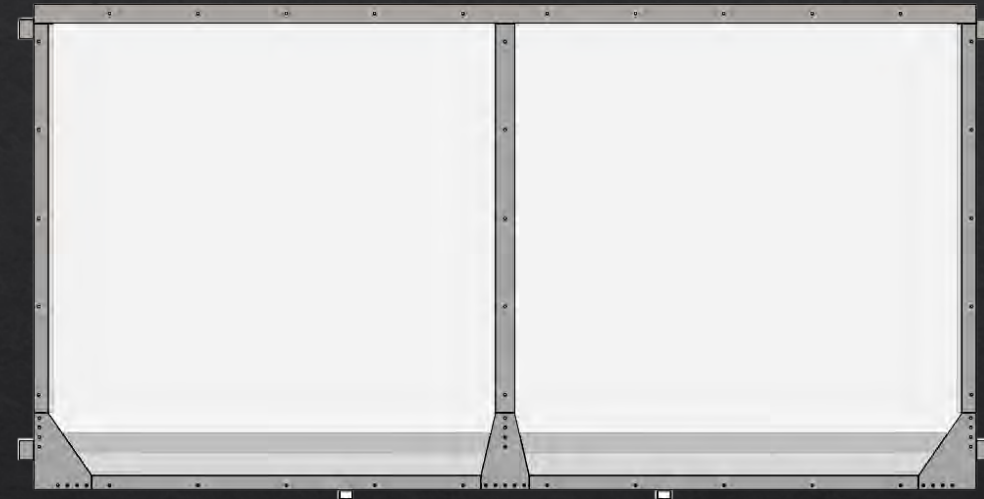
Beetleweight Arena

Beetleweight Combat Robotics Arena

- ◆ I was assigned to lead the development and construction of a new arena for 3-lb combat robots for the Purdue University combat robotics club
- ◆ Design requirements and constraints:
 - ◆ Large footprint for robot combat - 64 square foot minimum
 - ◆ Easy to assemble, transport and store as our competition venue and workspace are 4 miles apart
 - ◆ Quickly serviceable in case of damages throughout the tournament
 - ◆ High visibility from all angles
 - ◆ Double-walled polycarbonate panels with an air-gap to safely contain combat robots
 - ◆ Negative pressure ventilation system to extract fumes in case of battery fire
 - ◆ \$10,000 budget

Beetleweight Arena - Construction

- ◆ I came up with a design totaling \$7,000, saving \$3,000, that meets all the necessary requirements, while prioritizing serviceability and ease of assembly
- ◆ 8' long, 8' wide, 4' tall arena stands 3' above the ground providing ample space for combat at a comfortable viewing height
- ◆ The entire frame is assembled and disassembled without use of fasteners, enabling rapid setup
- ◆ Double-layered plywood floor provides a safe, durable, and smooth driving experience



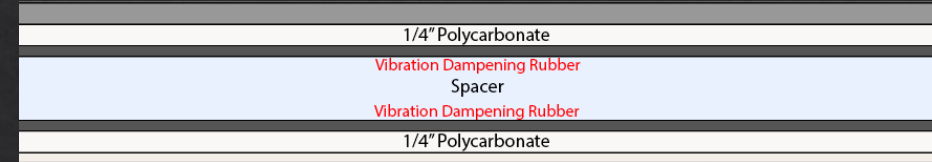
Each frame has a ~4'x8' footprint and can be transported flat-packed on a 4'x8' trailer



Frames assemble by being stacked at peg & tube joint at vertical bracket

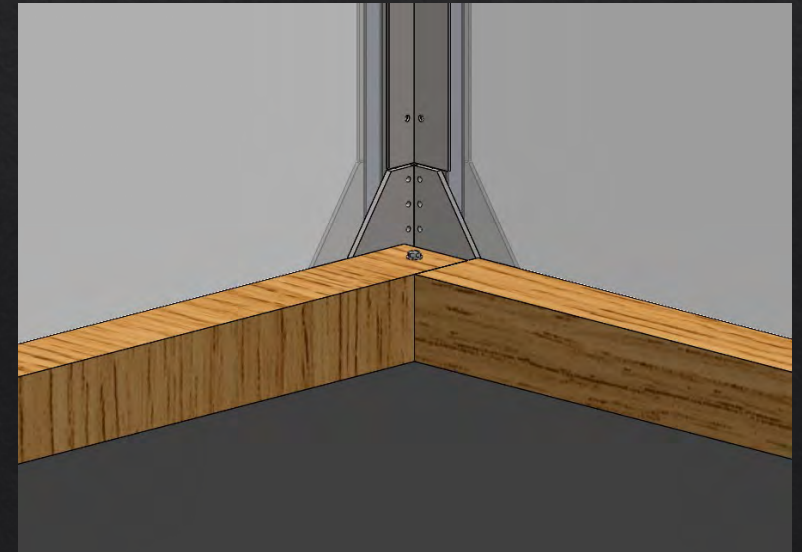
Beetleweight Arena - Safety

- ❖ Safety is the highest consideration in the design of this arena, guiding the decisions I made
- ❖ Double layered $\frac{1}{4}$ " polycarbonate panels have a 1" air-gap to minimize damage to the outer layer and reduce the chance of a breach
- ❖ An 800 cfm fan can pull out all air within the arena in ~20 seconds, extracting toxic LiPo battery fumes in the event of fire
- ❖ Sensored door lock changes the arena lights from white to red when the doors are unlocked



Beetleweight Arena - Serviceability

- ◆ Polycarbonate panels are inserted from the top and can be replaced in less than 10 minutes
- ◆ Polycarbonate panels bisected to save costs; replacing a 4x8 panel is much more expensive than a 4x4 panel
- ◆ Consumable wood kickplates limit gashes and scrapes from damaging polycarbonate
- ◆ Floor panels can be inserted through the door and replaced in 20 minutes



Arena - PDM

- ❖ We're using the Aras Innovator PDM system to collaborate with other club members on the design and share data for manufacturing
- ❖ I trained members on understanding and utilizing the PDM software
- ❖ Using PDM to link part drawings for manufacturing to their Part Documents

P0004039 ☆ □

Edit ↺ ↻ ↶ ↷ ↸ ↹ ⋮

Part

Part Number: P0004039 Revision: A State: Preliminary Assigned Team: [Raiside Robotics](#)

Name: Beetleweight Arena TLA Designated User: [Raiside Robotics](#)

Type: Assembly Unit: IN Make / Buy: Make Cost: Effective Date:

Long Description:

☐ Changes Pending Control Type:

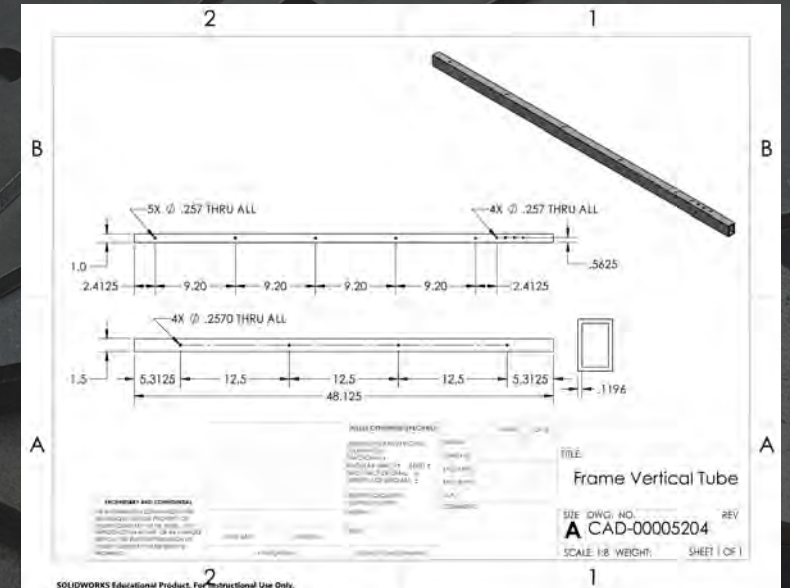
BOM Part NC **BOM Structure** Alternates AML Documents CAD Documents Analysis Containers Goals Changes Part Submission Warrant

3

Part Number	Re...	State	Sequence	Quantity	Claimed By	Name	Effectivity
⊞ P0003940-001	A	Preliminary	5	2		Beetleweight Arena Wall Frame	
⊞ P0004036	A	Preliminary	25	3		BWAR Rubber Strip	
⊞ P0004036-001	A	Preliminary	30	4		BWAR Rubber Strip	
⊞ P0004041	A	Preliminary	40	2		BWAR L-Channel	
⊞ P0003939-001	A	Preliminary	45	1		BWAR Door Panel VerticalTube	
⊞ P0004040-002	A	Preliminary	70	1		BWAR UHMW Retaining Plate	
⊞ P0004036-002	A	Preliminary	75	1		BWAR Rubber Strip	
⊞ P0004054	A	Preliminary	90	1		BWAR Tube Mirror	
⊞ P0003939-003	A	Preliminary	105	1		BWAR Door Panel VerticalTube	
⊞ P0004140	A	Preliminary	110	4		BWAR Wall Peg Holder	
⊞ P0004213	A	Preliminary	125	2		BWAR Tube Guide	
⊞ P0004212-001	A	Preliminary	130	2		BWAR Floor Brace Mount	

Beetleweight Arena - Manufacturing

- ❖ Considering the scale of this project, after completing the design, the next biggest challenge for me has been managing the arena's manufacturing
- ❖ Leading a team of 20 members in manufacturing over 40 unique parts
- ❖ Using part drawings to ensure consistent and accurate parts are made
- ❖ Utilizing CNC Mills, lathes, laser cutters, and welding to construct the steel frame



Beetleweight Arena – Build Progress

